



KG COLLEGE OF ARTS AND SCIENCE

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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

NLP-BASED EMOTION DETECTION AND MENTAL HEALTH SUPPORT

PROJECT REPORT

Submitted to Bharathiar University in partial fulfillment of the requirements for the award of the degree of

Bachelor of Science in Computer Technology

Submitted by

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Under the Supervision and Guidance of

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MARCH 2026

CERTIFICATE

CERTIFICATE

This is to certify that the project entitled **NLP-Based Emotion Detection and Mental Health Support** submitted to Bharathiar University in partial fulfillment for the award of the degree of Bachelor of Science in Computer Technology is a record of original work done by **C N Harini (2326K0014)** during the period of study in **KG College of Arts and Science** under the supervision of **Mr. S. Deepankumar, Assistant Professor**, Department of Computer Technology.

Place: Coimbatore

Date:

Signature of the Guide

Signature of the HoD

(College Seal)

Submitted for the Viva-Voce Examination held on _____

Internal Examiner

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DECLARATION

DECLARATION

I declare that the work which is being presented in this project report entitled **NLP-Based Emotion Detection and Mental Health Support**, submitted to the Department of Computer Technology, KG College of Arts and Science, Saravanampatti, Coimbatore for the award of the degree of Bachelor of Science in Computer Technology of the Bharathiar University, is an authentic record of my work carried out under the supervision and guidance of **Mr.S.Deepankumar M.C.A.,(Ph.D)**. I have not plagiarized or submitted the same work for the award of any other degree.

March 2026

Coimbatore

(C.N Harini)

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- C N Harini

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ABSTRACT

The NLP-Based Emotion Detection and Mental Health Support System is developed to provide accessible and immediate mental health assistance using advanced technologies such as Natural Language Processing (NLP), Sentiment Analysis, and Large Language Models (LLMs). Many individuals face difficulties accessing professional mental health support due to high costs, long waiting times, and social stigma. This system aims to address these challenges by offering an intelligent digital platform where users can express their emotions and receive supportive responses from an AI-based virtual psychologist.

The system analyzes user messages to detect emotions and understand their mental state. It uses a Dual-Layer Emotion Intelligence approach that combines lexicon-based emotion analysis with the contextual capabilities of the Groq AI engine. This combination helps accurately identify emotions such as stress, anxiety, sadness, or severe distress and provides appropriate responses and coping suggestions.

The platform also performs real-time risk assessment. If the system detects critical emotional conditions, it can generate alerts and support emergency intervention. Conversation history is stored to analyze emotional patterns and improve personalized responses.

The application is built using Python and Flask for the backend, SQLite for database management, and Groq API for intelligent language responses. The frontend is developed using HTML5, JavaScript, and Tailwind CSS, providing a responsive and user-friendly interface.

1.INTRODUCTION

Mental health challenges are increasing globally due to academic stress, work pressure, financial instability, and social isolation. Traditional mental healthcare systems often face limitations such as restricted operating hours, high consultation costs, social stigma, and long waiting periods. These barriers prevent many individuals from seeking timely psychological support. The NLP-Based Emotion Detection and Mental health support system is developed as a technological solution to address these limitations by offering a secure, automated, and continuously available mental health support platform.

The system leverages Artificial Intelligence technologies to analyze user conversations and detect emotional distress in real time. Using Natural Language Processing (NLP) and sentiment analysis, the platform evaluates textual inputs and assigns weighted emotion scores. These scores are processed through a risk assessment algorithm that determines the severity of the user's emotional condition. This automated mechanism reduces the chances of human error and ensures consistent evaluation of distress signals.

A key feature of the system is its Real-Time Emergency Response Mechanism. When high-risk phrases or suicidal ideation markers are detected, the system automatically triggers an emergency alert. The alert is logged in the database and made accessible through the administrator dashboard. In cases where users grant location access, geographical data is stored securely to assist emergency responders if necessary.

The system serves two primary user groups: general users seeking emotional support and administrators responsible for monitoring critical cases. Users can register anonymously, ensuring privacy and encouraging open communication. Administrators can monitor risk distribution, review alerts, and manage response protocols efficiently.

By combining automation, intelligent risk detection, secure data management, and user-friendly design, the system represents a modern, scalable solution to improve accessibility, efficiency, and safety in mental healthcare support.

1.1ABOUT THE PROJECT

The NLP-Based Emotion Detection and Mental health support project is an advanced mental healthcare management system designed to provide intelligent, accessible, and secure emotional support using Artificial Intelligence technologies. The primary goal of this project is to bridge the gap between individuals experiencing psychological distress and the availability of timely professional assistance. By integrating Natural Language Processing (NLP), Sentiment Analysis, and Large Language Models (LLMs), the system delivers empathetic, context-aware responses while continuously monitoring emotional risk levels.

The platform introduces a Dual-Layer Emotion Intelligence System that combines a lexicon-based emotion scoring engine with advanced AI-driven contextual understanding. This hybrid approach ensures accurate detection of emotional states ranging from mild stress to critical crisis conditions. The system automatically calculates a real-time risk score for each user message and categorizes it into Low, Medium, High, or Critical levels. If the risk score exceeds a predefined threshold, an emergency alert is generated and stored in the database for administrative review.

NLP-Based Emotion Detection and Mental health support is structured into multiple integrated modules, including User Interface, Patient Registration, NLP & Emotion Analysis, Chat Management, Emergency Alert, and Data Management. The backend is developed using Python and Flask, while SQLite and SQLAlchemy handle secure database operations. The frontend is built using HTML, JavaScript, and Tailwind CSS to provide a responsive and psychologically safe interface.

Overall, the project demonstrates how AI-driven automation can enhance mental healthcare accessibility, reduce manual workload, ensure consistent risk assessment, and improve crisis intervention efficiency.

1.2 SYSTEM SPECIFICATIONS

1.2.1 HARDWARE CONFIGURATION

- **Processor** : Intel Core i5 or AMD Ryzen 5 (or higher)
- **RAM** : 8 GB minimum (16 GB recommended)
- **Storage** : 512 GB
- **Network Interface** : Gigabit Ethernet or High-Speed Wi-Fi Adapter
- **Peripheral Devices** : Standard Keyboard and Mouse Webcams and Microphones

1.2.2 SOFTWARE SPECIFICATIONS

- **IDE** : Visual Studio Code
- **Frontend** : HTML5 , CSS3 / Tailwind CSS , JavaScript
- **Backend** : Python , Groq SDK
- **Database** : SQLite
- **Operating System** : Windows 10/11 or Linux
- **Browser** : Google Chrome, Mozilla Firefox, or Microsoft Edge (latest versions).

2. SYSTEM STUDY

2.1 EXISTING SYSTEM

The traditional systems for managing mental health care and patient support rely heavily on manual or face-to-face methods. These include in-person therapy sessions, crisis hotlines managed by human operators, and physical paper records for patient history. While these systems provide high-quality care, they lack the scalability, immediate accessibility, and continuous availability needed for modern healthcare environments.

In the current system, patient verification and initial triage depend on scheduling appointments or waiting on hold for a hotline operator. These processes are prone to delays and can be deter individuals in acute distress. Furthermore, the reliance on handwritten notes or disconnected electronic health records (EHRs) often leads to fragmentation of information, making it difficult to track a patient's emotional history over time or identify inconsistencies in their narrative.

2.1.1 DRAWBACKS

1. **Limited Availability:** Traditional therapy is often restricted to business hours (9-5), leaving patients without support during nights and weekends when crises often occur.
2. **High Cost:** Private therapy sessions are prohibitively expensive for many individuals, creating a socioeconomic barrier to mental healthcare.
3. **Stigma and Privacy:** The requirement to physically visit a clinic or speak to a human can induce anxiety and fear of judgment, preventing many from seeking help.
4. **Delayed Intervention:** Waiting lists for therapists can range from weeks to months, during which a patient's condition may deteriorate significantly.
5. **Subjective Triage:** Human assessment of risk can vary between practitioners; fatigue and burnout among hotline operators can lead to missed warning signs.
6. **Fragmented Data:** Lack of a centralized, continuous log of the patient's emotional state makes it hard to see patterns or triggers over time.

2.2 PROPOSED SYSTEM

The NLP-Based Emotion Detection and Mental Health Support system addresses the challenges of existing systems by introducing an integrated, automated, and secure platform. By combining modern AI technologies with user-centric design, the proposed system ensures accurate emotion tracking, immediate support, and seamless data management.

In this system, patient identification and interaction are achieved through secure digital sessions and NLP-based context recognition. This ensures a high level of accuracy and security. Conversations are digitized and stored in a centralized database (mental_health.db), allowing for real-time updates and easy access (by the AI). The system automates risk assessment and facilitates location-based emergency alerts, enhancing convenience and safety for patients and staff alike.

2.2.1 FEATURES

1. **Dual-Layer Emotion Detection:** Combines a rule-based lexicon scorer for immediate keyword flagging with a Generative AI model for contextual nuance, ensuring reliable identification of distress.
2. **24/7 Availability:** The AI psychologist is always online, providing immediate responses to users regardless of the time or day.
3. **Real-time Risk Scoring:** Automatically calculates a risk score (0-100) for every user message, updating the user's status instantly.
4. **Emergency Alert System:** Automatically triggers an alert record if the risk score exceeds a critical threshold (e.g., 70), capturing location data for potential intervention.
5. **Secure & Anonymous:** Users can interact without revealing their full identity, encouraging honesty and reducing the fear of stigma.
6. **Integrated Platform:** Facilitates seamless communication between the user interface, the analysis engine, and the database.
7. **Data Security:** Protects sensitive patient information using secure session management and database isolation.
8. **User-friendly Interface:** Simplifies navigation for users in distress, requiring no training to start a conversation.
9. **Analytics and Reporting:** Provides structured data on emotional trends, helping to understand user needs better.

3. SYSTEM DESIGN AND ENHANCEMENT

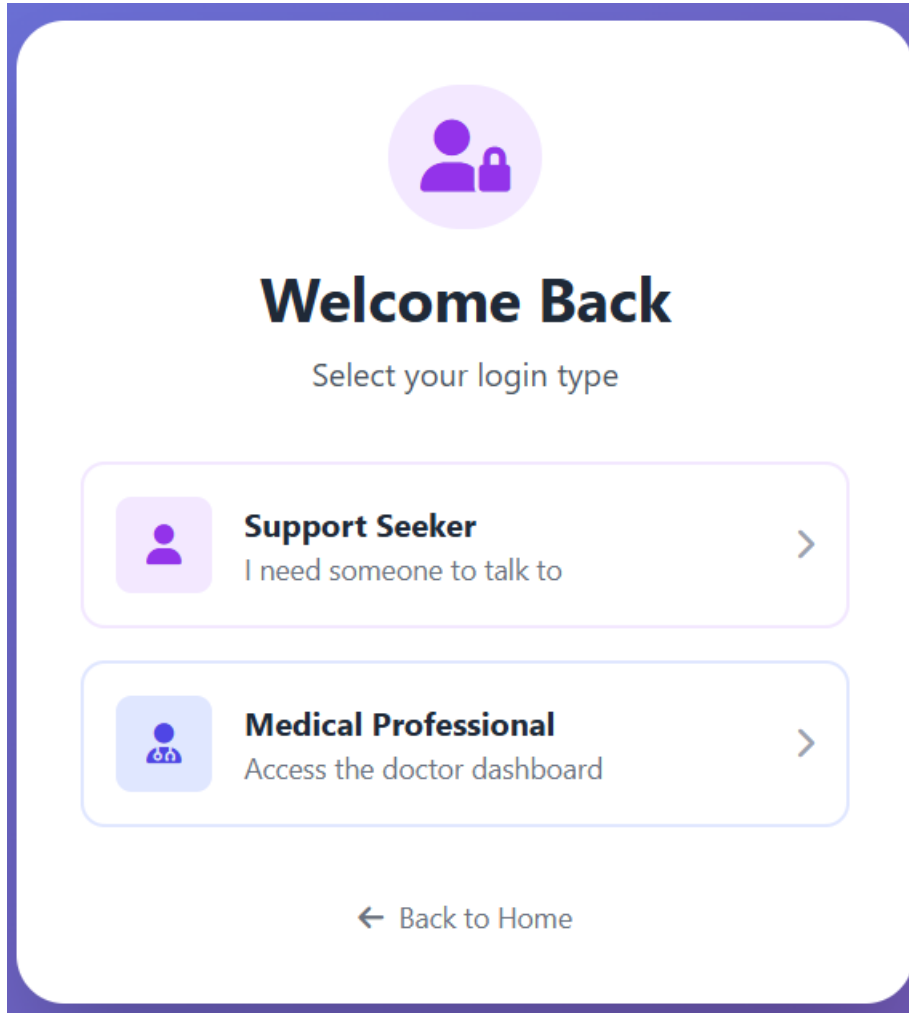
3.1 FILE DESIGN

The NLP-Based Emotion Detection and Mental Health Support system is designed with a structured file management approach to ensure seamless data organization, storage, and retrieval. At its core, the project directory is divided into backend and frontend folders to separate concerns.

- **Backend Directory:** Contains the application logic.
 - **main.py:** The entry point of the Flask application, defining API routes and server configuration.
 - **database.py:** Defines the SQLAlchemy models (User, ChatHistory, EmergencyAlert) and the DatabaseManager class for abstraction.
 - **nlp_engine.py:** Contains the EmotionScorer class and the logic for calculating risk scores based on the lexicon.json.
 - **groq_client.py:** Manages the connection to the Groq API, handling prompt engineering and response retrieval.
 - **lexicon.json:** A structured data file containing keywords mapped to emotions and their severity weights.
 - **mental_health.db:** The SQLite database file storing persistent records.
- **Frontend Directory:** Contains the presentation layer.
 - **index.html:** The landing page with the “Get Started” call to action.
 - **login.html / register.html:** User authentication forms (simulated or functional).
 - **chat.html:** The main interface for the chat session.
 - **dashboard.html:** The admin view for monitoring alerts.
 - **js/:** Contains JavaScript files (chat.js, dashboard.js) that handle API calls and DOM manipulation.

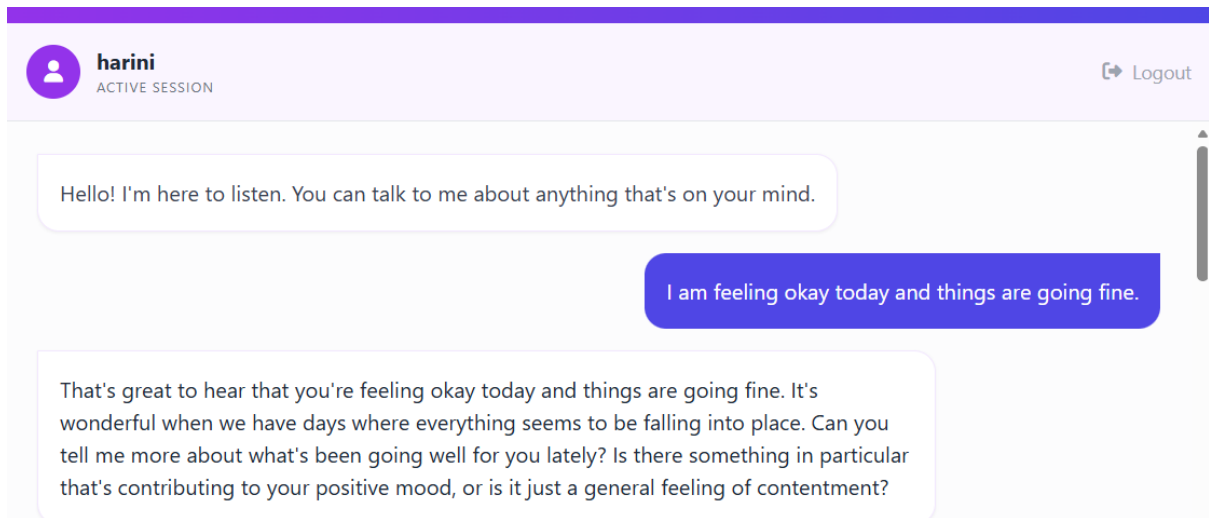
This structured approach enhances maintainability, reducing code redundancy and improving overall efficiency in development and debugging. Configuration files like .env store API credentials and system parameters, ensuring security by keeping sensitive keys out of the codebase.

3.2 INPUT DESIGN



This screen shows the login selection page of the MindCare AI system. Users can choose to log in as a Support Seeker to get help or as a Medical Professional to access the doctor dashboard.

3.3 OUTPUT DESIGN



This output screen shows the real-time interaction between the user and the AI psychologist. The system analyzes the user's message, determines a low-risk emotional state, and generates an empathetic response without triggering any emergency alert.

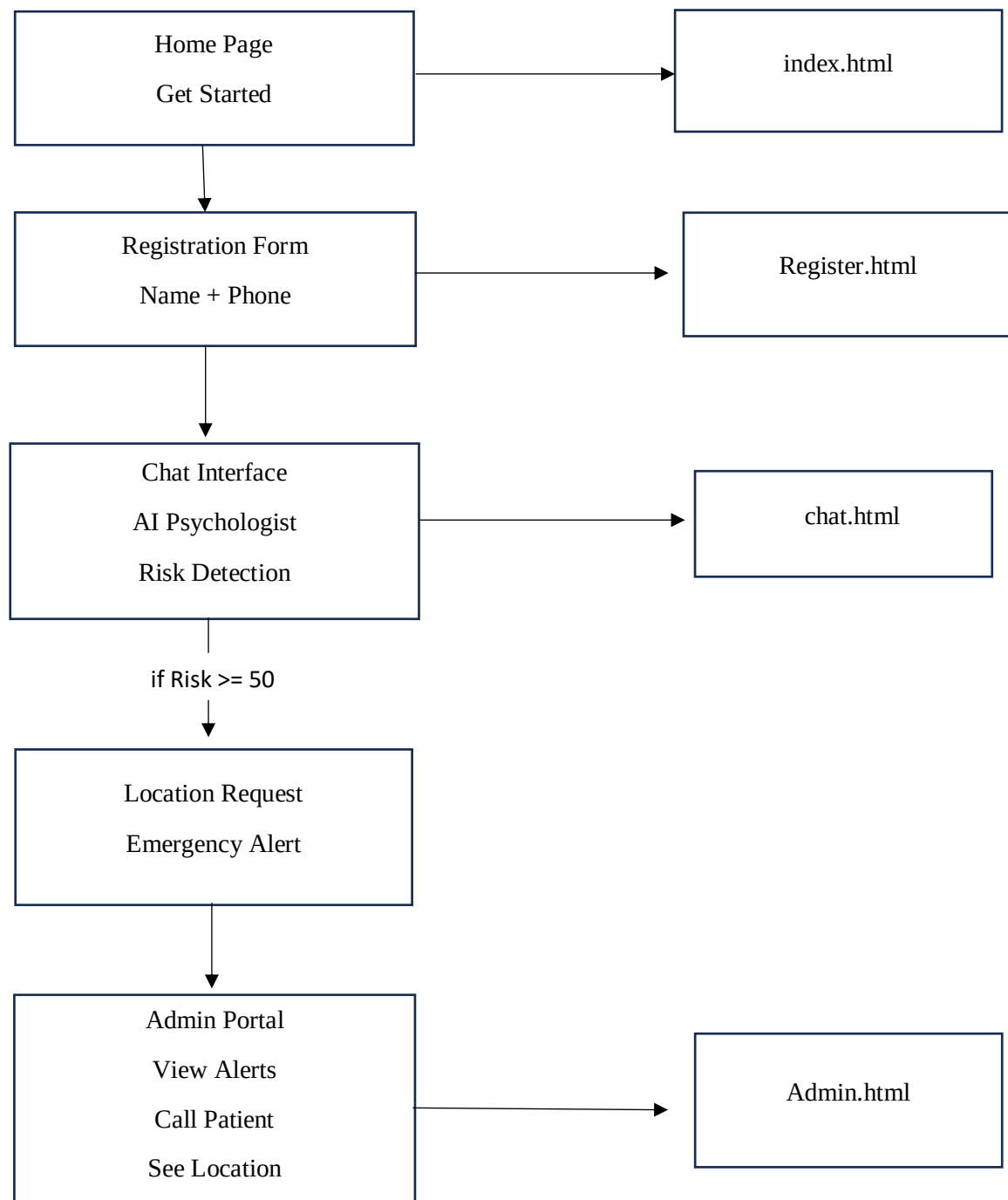
3.4 DATABASE DESIGN

The database structure of the system is designed for optimal performance and data integrity. The primary database, `mental_health.db`, is structured into multiple tables using SQLAlchemy's declarative base.

- **Normalization:** The database is normalized to reduce redundancy. User details are stored once in the `users` table, while multiple chat messages are stored in `chat_history` linked by `user_id`.
- **Indexing:** Primary keys (UUIDs) are indexed for fast retrieval.
- **Data Types:** Efficient data types are chosen (e.g., Float for coordinates, Boolean for emergency flags, DateTime for timestamps).

The system employs data validation techniques, ensuring that every entry is cross-checked before being added to the database. These techniques prevent duplicate patient records and inconsistencies.

3.5 CODE DESIGN



The user starts on the home page, registers with their name and phone, and then chats with an AI psychologist. The system analyzes the conversation for risk, and if the risk level is high (≥ 50), it sends an alert with the user's location to the admin portal for immediate action.

3.6 SYSTEM MODULES

The system modules phase involves designing, implementing, and integrating various components to ensure efficient and reliable patient identification and mental health tracking. The Tracker follows a modular development approach, ensuring scalability, maintainability, and security. Each module is designed to handle a specific function, allowing for seamless communication between different parts of the system.

MODULES

1. User Interface
2. Patient Registration
3. NLP & Emotion Analysis
4. Chat Management
5. Emergency Alert

3.6.1 MODULE DESCRIPTION

USER INTERFACE MODULE

The User Interface Module is designed using HTML and Tailwind CSS to provide an interactive and responsive web interface for users. It allows smooth navigation between Home, Chat, and Resources sections. The module supports real-time chat display using a scrolling message container and applies custom CSS for a professional and calming user experience. Asynchronous JavaScript using the Fetch API ensures non-blocking communication with the backend server.

PATIENT REGISTRATION MODULE

The Patient Registration Module handles the collection and storage of patient information in a secure and anonymous manner. It uses SQLAlchemy to create and manage user records in the database. Unique patient identification is achieved through UUID generation, ensuring privacy. Input forms are provided to capture optional demographic information without mandating personal details.

NLP & EMOTION ANALYSIS MODULE

The NLP and Emotion Analysis Module identifies the emotional state of the user through text-based analysis. It uses the EmotionScorer class to compare user input with predefined emotional keywords stored in the lexicon.json file. Regular expressions are applied for text preprocessing, and weighted arithmetic calculations are used to compute emotion scores and overall mental health risk levels.

CHAT MANAGEMENT MODULE

The Chat Management Module controls user-system interactions and ensures meaningful conversations. It utilizes the GroqClient to generate intelligent and context-aware responses. The module maintains conversation continuity by storing chat history in a conversation store and logs all interactions into the chat_history database table for monitoring and analysis.

EMERGENCY ALERT

The Emergency Alert Module is responsible for detecting critical mental health conditions and triggering emergency notifications. It continuously monitors the calculated risk score after each interaction. When the risk score exceeds a predefined threshold, the module generates alert records in the emergency_alerts table and exposes them through an API endpoint for dashboard access.

DATA MANAGEMENT MODULE

The Data Management Module ensures efficient and secure handling of patient records and chat data. It uses SQLAlchemy sessions for database operations such as data retrieval and storage. Exception handling mechanisms are implemented to manage errors and ensure safe rollbacks. Patient data is secured through unique identifier-based authentication to maintain confidentiality and integrity.

3.7 REPORT

Mental health issues are becoming increasingly common due to stress, academic pressure, work-related challenges, and social factors. Many individuals hesitate to seek professional help because of stigma, lack of awareness, or limited access to mental health services. To address this issue, this project proposes an NLP-Based Emotion Detection and Mental Health Support System that provides timely emotional assistance using Artificial Intelligence techniques.

The proposed system uses Natural Language Processing (NLP) to analyze user text inputs and identify emotional states such as happiness, sadness, anger, anxiety, and suicidal tendencies. Users can freely express their feelings through text, and the system processes this input to detect emotions and assess the severity of mental distress. Based on the detected emotions, the system categorizes the user's condition into different risk levels such as low, medium, high, or critical.

Once the risk level is identified, the system generates empathetic and supportive responses to help the user cope with their emotional state. In critical situations, where suicidal or self-harm intent is detected, the system automatically triggers an emergency alert mechanism to notify administrators or support personnel for immediate action. All user interactions and emotion analysis results are securely stored in the database for monitoring and future reference.

The system is designed using a modular architecture that includes an emotion detection module, risk assessment module, response generation module, database module, and alert module. Testing techniques such as unit testing, integration testing, system testing, user acceptance testing, and performance testing are conducted to ensure accuracy, reliability, and efficiency of the system.

Overall, the NLP-Based Emotion Detection and Mental Health Support System provides an accessible, confidential, and real-time solution for mental health assistance. While it does not replace professional therapy, it serves as an effective support tool for early detection of emotional distress and timely intervention.

4. TESTING AND IMPLEMENTATION

Testing and implementation play a crucial role in ensuring the systems functions efficiently and securely. This phase involves evaluating system performance, verifying accuracy, and integrating the software within a real-world healthcare setting context. The testing strategy follows a structured approach, including unit testing, integration testing, and user acceptance testing to ensure the system meets functional and non-functional requirements.

4.1 TESTING METHODS

- Unit Testing
- Integration Testing
- System Testing
- User Acceptance Testing (UAT)
- Performance Testing

UNIT TESTING

Unit testing is performed to verify the correct functioning of individual system components. Modules such as emotion scoring, API endpoints, and database operations are tested separately. Test cases include checking EmotionScorer accuracy with phrases like “I want to die,” validating proper JSON responses from Flask routes, and ensuring user data is correctly stored and retrieved from the database.

TABLE 4.1: UNIT TESTING

Test Case ID	Module	Test Description	Input	Expected Output	Actual Output	Status
UT01	EmotionScorer	Detect sadness keywords	"I feel very sad and lonely"	Sadness score > 0, Medium risk	As expected	Pass
UT02	EmotionScorer	Detect suicidal intent	"I want to die"	Risk score > 70, Emergency = True	As expected	Pass
UT03	Chat API	Validate empty message	"" (empty string)	Error message returned	As expected	Pass
UT04	Database	Store chat history	Valid message	Record saved in chat_history table	As expected	Pass
UT05	Alert Module	Generate emergency alert	Risk score = 85	Entry created in emergency_alerts	As expected	Pass

INTEGRATION TESTING

Integration testing ensures that different modules work correctly when combined. It verifies the interaction between `main.py`, `nlp_engine.py`, and `groq_client.py`, ensuring smooth data flow. Tests include checking that a user message triggers the NLP engine, saves data to the database, and returns an AI response, confirming emergency alerts are created for high-risk scores, and verifying that the frontend chat window correctly displays messages from the backend.

TABLE 4.2: INTEGRATION TESTING

Test Case ID	Modules Involved	Test Description	Input	Expected Result	Status
IT01	UI → Backend → NLP	User sends emotional message	“I feel hopeless”	NLP processes text and returns response	Pass
IT02	NLP → Database	Emotion score storage	Emotional text	Emotion & risk score saved	Pass
IT03	NLP → Alert → DB	High-risk detection flow	“I want to kill myself”	Alert generated & stored	Pass
IT04	Backend → UI	AI response rendering	Normal user text	AI reply shown in chat UI	Pass
IT05	Backend → Dashboard	Alert visibility	Emergency triggered	Alert visible in admin dashboard	Pass

SYSTEM TESTING

System testing verifies that the entire system works correctly as a whole. It checks the integration of the user interface, NLP emotion analysis, AI response generation, database storage, and emergency alert system. The test ensures accurate risk detection and proper alert triggering.

TABLE 4.3: SYSTEM TESTING

Test Case ID	Feature	Test Scenario	Expected Outcome	Status
ST01	User Registration	Register new user	User ID generated successfully	Pass
ST02	Chat Functionality	Continuous conversation	Context maintained	Pass
ST03	Risk Classification	Different emotions	Correct risk category shown	Pass
ST04	Location Capture	User allows location	Latitude & longitude stored	Pass
ST05	Emergency Handling	Critical emotion detected	Emergency alert triggered	Pass

USER ACCEPTANCE TESTING

User Acceptance Testing ensures that the system is easy to use and meets user expectations. Users can interact anonymously, receive empathetic AI responses, and administrators can view emergency alerts clearly. The system was accepted as user-friendly and effective.

TABLE 4.4: USER ACCEPTANCE TESTING

Test Case ID	User Type	Scenario	Expected Result	Status
UAT01	General User	Chat anonymously	No personal data required	Pass
UAT02	Distressed User	Express suicidal thoughts	Immediate supportive response	Pass
UAT03	Admin	View emergency alerts	Alerts listed clearly	Pass
UAT04	Admin	Resolve alert	Status updated to “Resolved”	Pass
UAT05	Mobile User	Use system on phone	Responsive UI works	Pass

PERFORMANCE TESTING

Performance testing evaluates the speed, responsiveness, and stability of the system. It provides real-time responses, handles multiple users efficiently, and triggers alerts without delay, ensuring reliable performance.

TABLE 4.5: PERFORMANCE TESTING

Test Case ID	Parameter	Expected Value	Observed Result	Status
PT01	API Response Time	< 3 seconds	~1.5 seconds	Pass
PT02	Concurrent Users	20+ users	No crash	Pass
PT03	Database Write	Real-time logging	Successful	Pass

4.1.1 SYSTEM IMPLEMENTATION

1. Deployment Environment

The NLP-Based Emotion Detection and Mental Health Support system is designed for deployment on local servers or cloud-based platforms (like AWS, Heroku, or Render), ensuring real-time accessibility and secure data management. The implementation phase includes the following key steps:

- **Configuring database storage:** The system establishes a structured SQLite database to store patient records and history. The data file is secured with file-system permissions.
- **Setting up API integrations:** The system integrates Groq API keys into the environment variables (.env) to enable AI functionality.
- **Environment Setup:** Installing dependencies via requirements.txt (Flask, SQLAlchemy, etc.) in a virtual environment.

The deployment environment is tested rigorously to ensure scalability, security, and seamless interaction between different modules.

2. Training and Documentation

To ensure smooth adoption of the system, comprehensive training sessions (simulated via this documentation) are provided for system administrators. These sessions focus on essential functionalities, including:

- **Dashboard Monitoring:** A walkthrough on how to view and manage emergency alerts.
- **System Maintenance:** Guidance on backing up the mental_health.db file.
- **Privacy Protocols:** Training on how to handle sensitive user data ethically.

4.1.2 SYSTEM ENHANCEMENT

The NLP-Based Emotion Detection and Mental Health Support System can be enhanced further to improve its accuracy, scalability, and overall user experience. The current system mainly focuses on text-based emotion detection using Natural Language Processing techniques. As a major enhancement, voice-based emotion detection can be incorporated. By analyzing speech features such as tone, pitch, speed, and pauses, the system can identify emotions more precisely, especially in cases where users find it difficult to express feelings through text alone.

Another important enhancement is the integration of multilingual support. By allowing users to interact in multiple languages, the system can become accessible to a broader population across different regions and cultures. This enhancement will significantly increase usability and adoption of the system in real-world environments.

The system can also be improved by incorporating advanced machine learning and deep learning models, such as transformer-based language models and emotion classification networks. These models can learn from large datasets and continuously improve emotion detection accuracy and risk prediction over time. This will reduce false predictions and enhance the reliability of the system.

Additionally, the system can be enhanced by integrating real-time professional mental health support, such as live chat or call connectivity with counselors and emergency helplines during critical situations. This ensures immediate human intervention when required. The development of mobile and web-based applications will further improve accessibility, allowing users to seek support anytime and anywhere.

With these enhancements, the system can evolve into a more intelligent, adaptive, and comprehensive mental health support platform capable of addressing complex emotional needs eff

5. CONCLUSION

The NLP-Based Emotion Detection and Mental Health Support system has successfully demonstrated its potential to enhance efficiency, security, and accuracy in mental health support and crisis intervention. By integrating AI-driven emotion recognition and real-time conversation analysis, the system ensures that patients receive immediate attention and that high-risk cases are identified instantly. This automated approach significantly reduces the barrier to entry for therapy, prevents unauthorized access to sensitive records, and streamlines the entire mental health triage process. The combination of automation and digital record-keeping significantly enhances patient safety by minimizing human error and ensuring that the right intervention reaches the right individual at the right time.

One of the key strengths of the system is its ability to automate key processes, such as risk scoring and alert generation, thereby reducing the workload on human crisis responders. Traditional manual methods often lead to inefficiencies, mismanagement, and missed warning signs. By implementing this technology, healthcare providers can focus more on delivering clinical care rather than being burdened with administrative tasks. Furthermore, the system's user-friendly interface, powered by Tailwind and Flask, ensures a seamless experience for both healthcare providers and patients.

The modular and scalable architecture of the system makes it highly adaptable for future advancements. Features such as cloud-based data storage, voice-to-text integration, and AI-driven predictive analytics for long-term mental health monitoring can be integrated into future versions to further enhance its capabilities. Moreover, additional security measures such as end-to-end encryption and multi-factor authentication can ensure compliance with healthcare regulations while safeguarding sensitive medical information.

FUTURE ENHANCEMENT

The NLP-Based Emotion Detection and Mental Health Support System can be extended in the future to provide more advanced, accurate, and comprehensive mental health support. One of the major future enhancements is the integration of voice-based emotion detection. By analyzing speech features such as tone, pitch, and speaking rate, the system can identify emotions more effectively, especially for users who are unable to express their feelings clearly through text.

Another important enhancement is the implementation of face recognition and facial emotion detection. By analyzing facial expressions through a camera, the system can identify emotions such as happiness, sadness, anger, or stress in real time. This feature will allow the system to understand the user's emotional state more accurately and provide better support, especially when verbal or textual communication is limited.

Another significant enhancement is the implementation of multilingual support, which will allow users to interact with the system in different regional and international languages. This will increase accessibility and ensure inclusivity for users from diverse linguistic backgrounds. The system can also be improved by incorporating advanced machine learning and deep learning models, such as transformer-based language models, to enhance emotion classification accuracy and reduce false predictions. In future versions, the system can be integrated with professional mental health services, including live chat or appointment scheduling with certified counselors and psychologists. During critical situations, automatic integration with emergency helplines and crisis intervention services can ensure immediate human support.

Additionally, the development of mobile and web-based applications will make the system easily accessible anytime and anywhere. Features such as emotion trend analysis, personalized mental health recommendations, and secure cloud-based data storage can further enhance the effectiveness of the system. With these future enhancements, the system can evolve into a scalable, intelligent, and reliable mental health support platform capable of addressing complex emotional and psychological needs.

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APPENDICES

A. DATA FLOW DIAGRAM

A Data Flow Diagram (DFD) is a diagram that describes the flow of data and the processes that change data throughout a system. It's a structured analysis and design tool that can be used for flowcharting in place of or in association with information. Oriented and process oriented system flowcharts. Four basic symbols are used to construct data flow diagrams. They are symbols that represent data source, data flows, and data transformations and data storage. The points at which data are transformed are represented by enclosed figures, usually circles, which are called nodes.

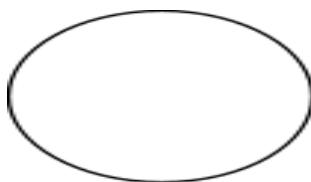
Data Flow Diagram Symbols



- Destination of the data



- Shows Data Flow

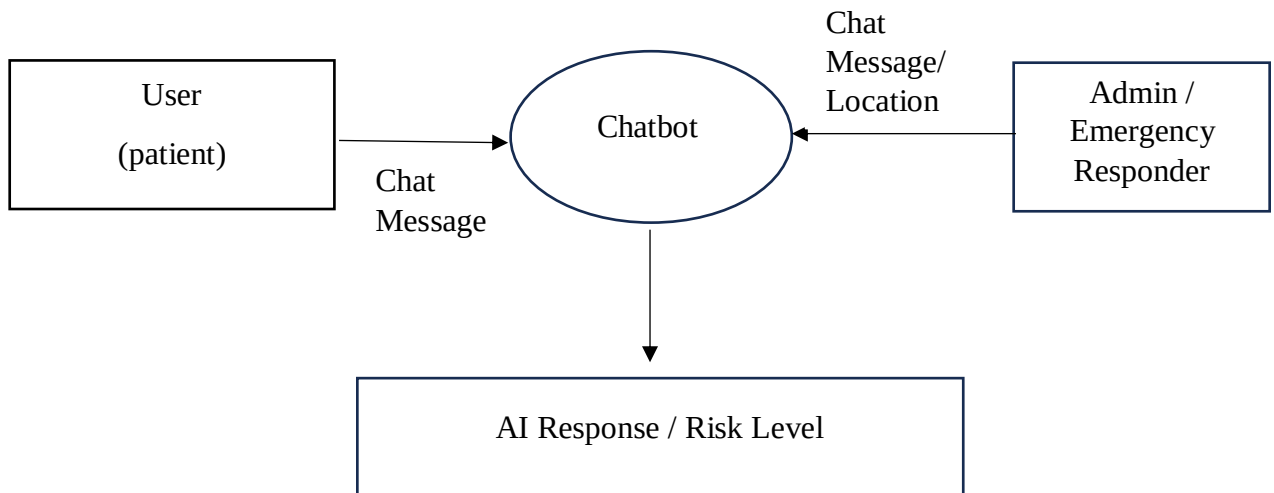


- Represent a process

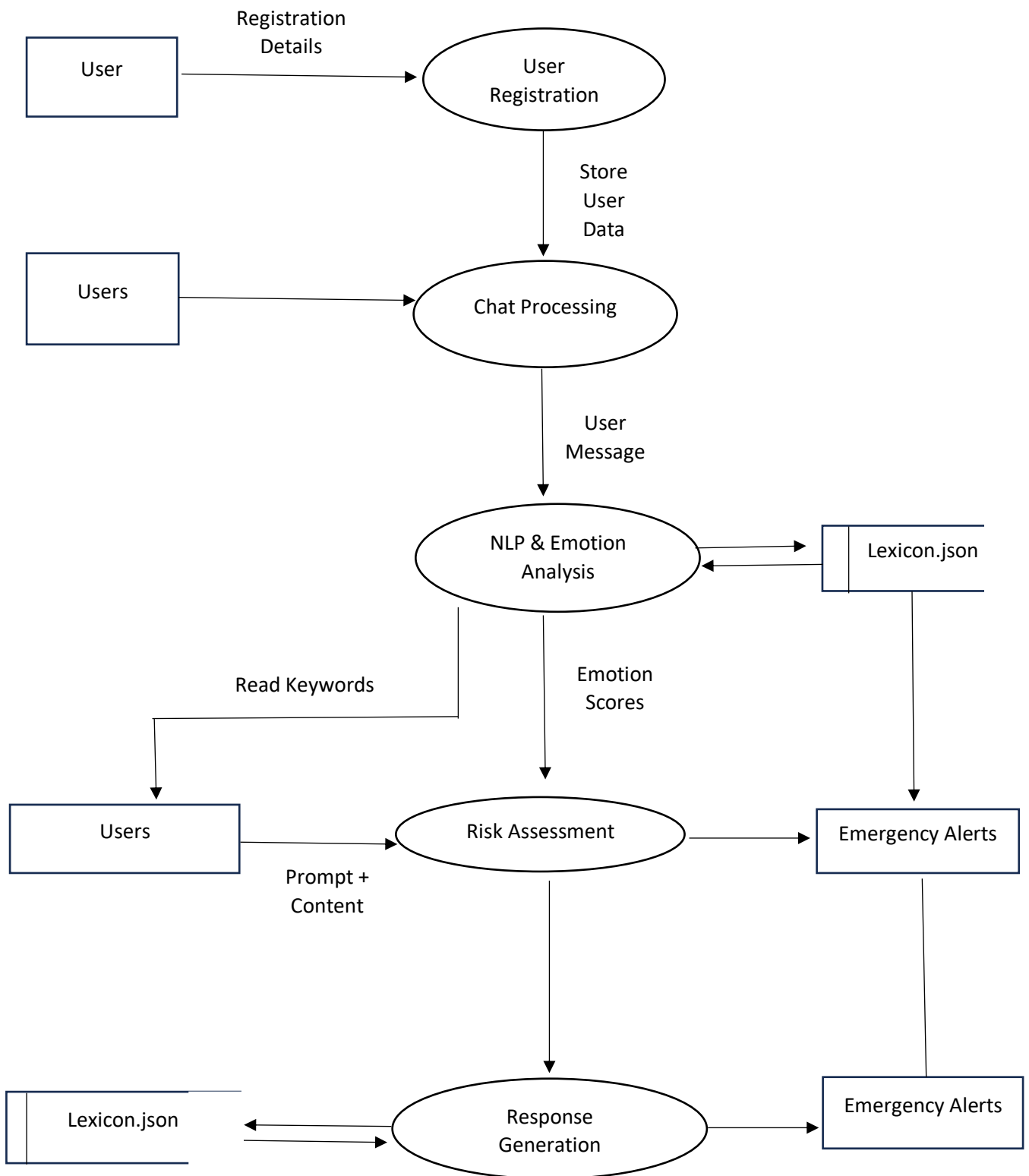


- Shows data source

LEVEL 0



LEVEL 1



B. TABLE DESIGN

TABLE NAME: USERS

PRIMARY KEY: ID

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier for patient (UUID)	PRIMARY KEY, NOT NULL
name	VARCHAR(100)	Full name of the patient (Optional)	NOT NULL
phone	VARCHAR(20)	Contact phone number	NOT NULL
location	VARCHAR(200)	Human-readable address/location	NOT NULL
latitude	FLOAT	GPS Latitude	NOT NULL
longitude	FLOAT	GPS Longitude	NOT NULL
created_at	DATETIME	Timestamp of registration	DEFAULT CURRENT_TIMESTAMP

TABLE NAME: CHAT_HISTORY

PRIMARY KEY: id

FOREIGN KEY: user_id

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier for the message	PRIMARY KEY, NOT NULL
user_id	VARCHAR(36)	Link to the users table	FOREIGN KEY
message	TEXT	The content of the user's message	NOT NULL
emotion_scores	JSON	Serialized dictionary of emotion scores	NOT NULL
risk_score	FLOAT	Calculated risk level (0-100)	NOT NULL
is_emergency	BOOLEAN	Flag for high-risk messages	DEFAULT FALSE
created_at	DATETIME	Timestamp of the message	DEFAULT CURRENT_TIMESTAMP

TABLE NAME: EMERGENCY_ALERTS

PRIMARY KEY: id

FOREIGN KEY: user_id

Field Name	Data Type	Description	Constraints
id	VARCHAR(36)	Unique identifier for the alert	PRIMARY KEY, NOT NULL
user_id	VARCHAR(36)	Link to the users table	FOREIGN KEY (users.id)
risk_score	FLOAT	The score that triggered the alert	NOT NULL
status	VARCHAR(20)	Status of the alert (active, resolved)	DEFAULT 'active'
created_at	DATETIME	Timestamp of the alert	DEFAULT CURRENT_TIMESTAMP

C. SAMPLE CODING

Emotion Detection using NLP

```
def calculate_emotion_scores(self, text: str) -> Dict[str, float]:  
    processed_text = self.preprocess_text(text)  
    words = processed_text.split()  
    scores = {}  
  
    for emotion, data in self.lexicon.items():  
        keywords = data['keywords']  
        weight = data['weight']  
        matches = 0  
  
        for keyword in keywords:  
            if keyword in processed_text:  
                matches += 1  
  
        raw_score = matches * weight * 10  
        scores[emotion] = min(raw_score, 100)  
  
    return scores
```

Weighted Risk Calculation

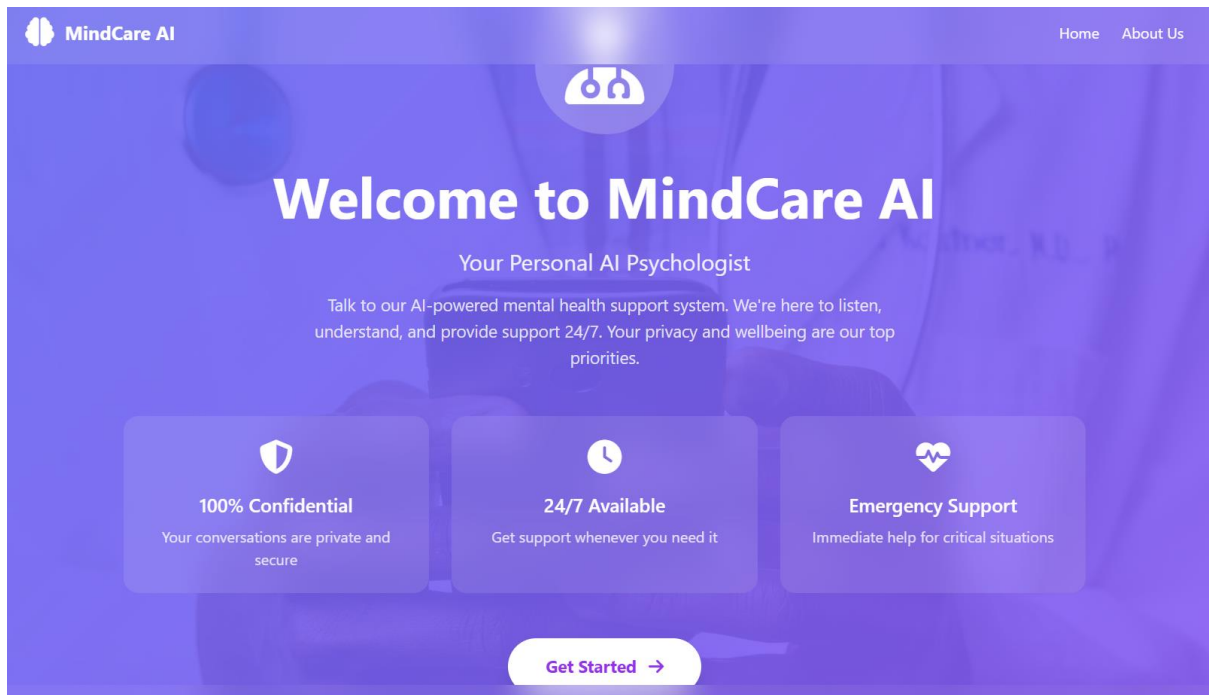
```
def calculate_risk_score(self, emotion_scores):  
    critical = ['suicidal', 'self_harm', 'hopelessness']  
    critical_max = max([emotion_scores.get(c, 0) for c in critical])  
  
    if critical_max >= 80:  
        return critical_max  
  
    final_score = critical_max * 0.7  
    return min(final_score, 100)
```

Risk Level Classification

```
def get_risk_level(self, risk_score):  
    if risk_score >= 70:  
        return "Critical"  
    elif risk_score >= 50:  
        return "High"  
    elif risk_score >= 30:  
        return "Medium"  
    else:  
        return "Low"
```

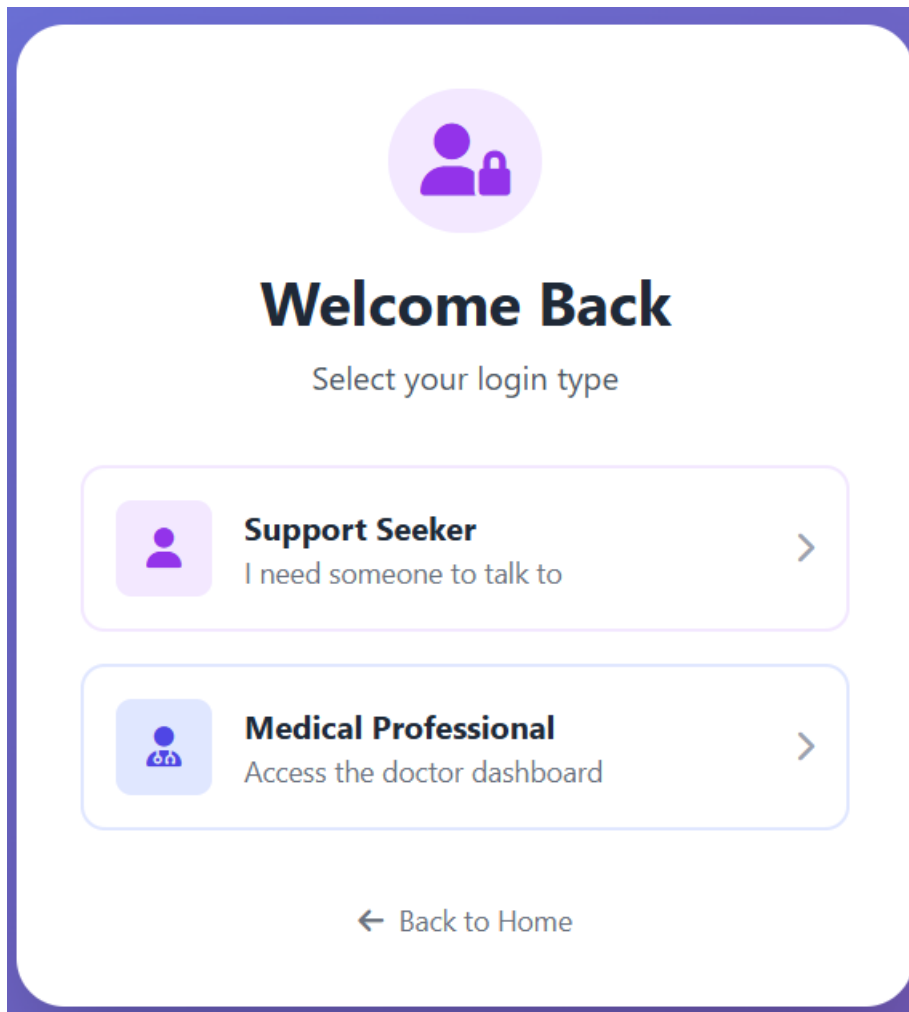
D. SAMPLE INPUT

LANDINGPAGE



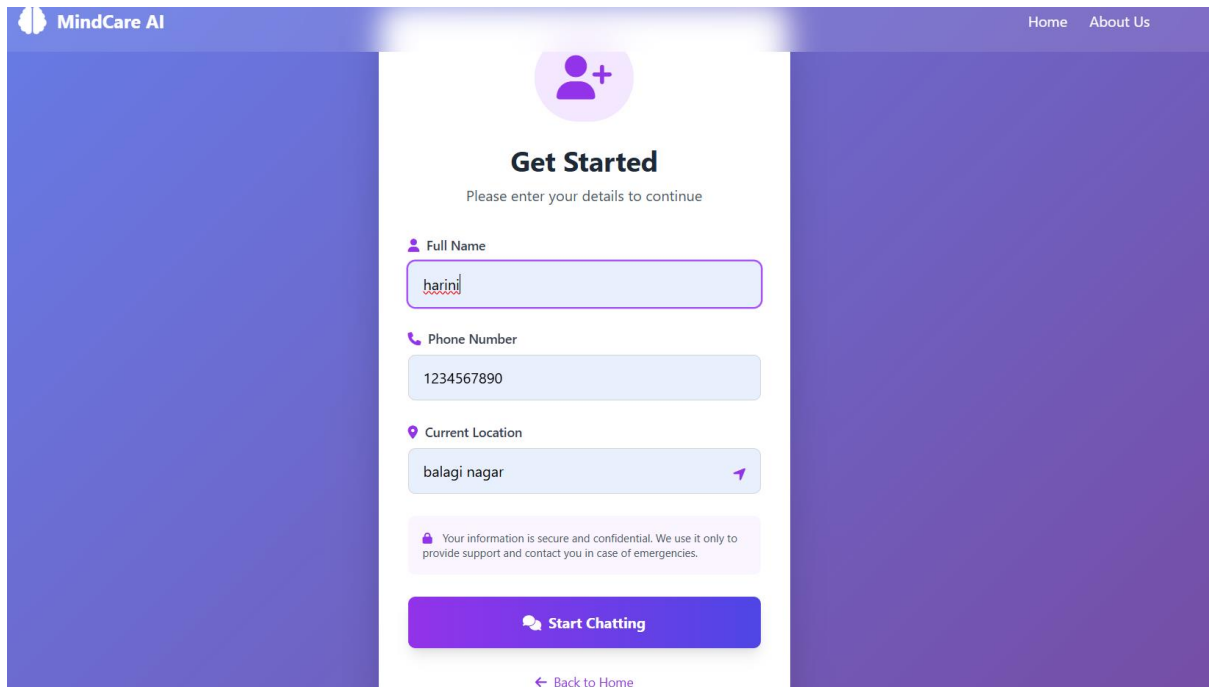
This screen displays the home page of the MindCare AI system, welcoming users to the platform. It highlights features such as confidential conversations, 24/7 support, and emergency mental health assistance.

USER / DOCTOR LOGIN



This screen shows the login selection page of the MindCare AI system. Users can choose to log in as a Support Seeker to get help or as a Medical Professional to access the doctor dashboard.

USER REGISTRATION



The image shows a user registration form for MindCare AI. The form is centered on a white background with a purple gradient border. At the top, there is a purple header with the MindCare AI logo on the left and 'Home' and 'About Us' links on the right. Below the header, there is a purple circle with a white person icon and a plus sign. The main heading is 'Get Started' in bold black text, followed by the instruction 'Please enter your details to continue'. The form contains three input fields: 'Full Name' with the value 'harini', 'Phone Number' with the value '1234567890', and 'Current Location' with the value 'balagi nagar'. Below these fields is a purple box with a lock icon and the text 'Your information is secure and confidential. We use it only to provide support and contact you in case of emergencies.' At the bottom, there is a large purple button with a white speech bubble icon and the text 'Start Chatting'. Below the button is a link that says '← Back to Home'.

MindCare AI Home About Us

Get Started
Please enter your details to continue

Full Name
harini

Phone Number
1234567890

Current Location
balagi nagar

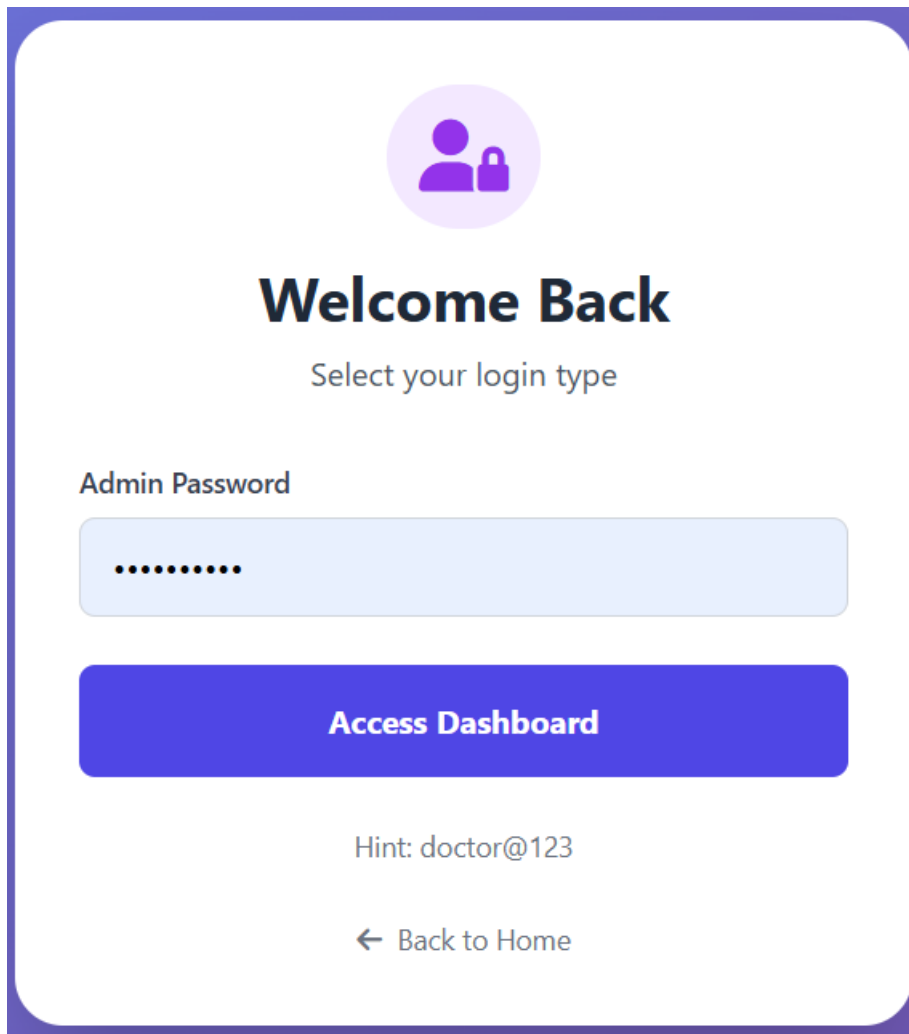
Your information is secure and confidential. We use it only to provide support and contact you in case of emergencies.

Start Chatting

← Back to Home

This screen allows users to enter personal details such as name, phone number, and location to begin using the MindCare AI system. After filling the details, users can click “Start Chatting” to start the support conversation.

ADMIN LOGIN



The image shows a login interface for medical professionals. At the top, there is a purple circular icon containing a white silhouette of a person and a padlock. Below this icon, the text "Welcome Back" is displayed in a large, bold, black font. Underneath, the instruction "Select your login type" is shown in a smaller, gray font. A label "Admin Password" is positioned above a light blue rectangular input field. Inside the input field, there are ten black dots representing a masked password. Below the input field is a solid blue button with the white text "Access Dashboard". At the bottom of the form, a hint "Hint: doctor@123" is provided in gray. Finally, there is a link "← Back to Home" in gray at the very bottom.

Welcome Back

Select your login type

Admin Password

.....

Access Dashboard

Hint: doctor@123

← Back to Home

This screen is used by medical professionals to access the dashboard by entering the admin password. After authentication, they can view and manage user support information.

LOW RISK

**harini**
ACTIVE SESSION

Logout

Hello! I'm here to listen. You can talk to me about anything that's on your mind.

I am feeling okay today and things are going fine.

MEDIUM RISK

**harini**
ACTIVE SESSION

Logout

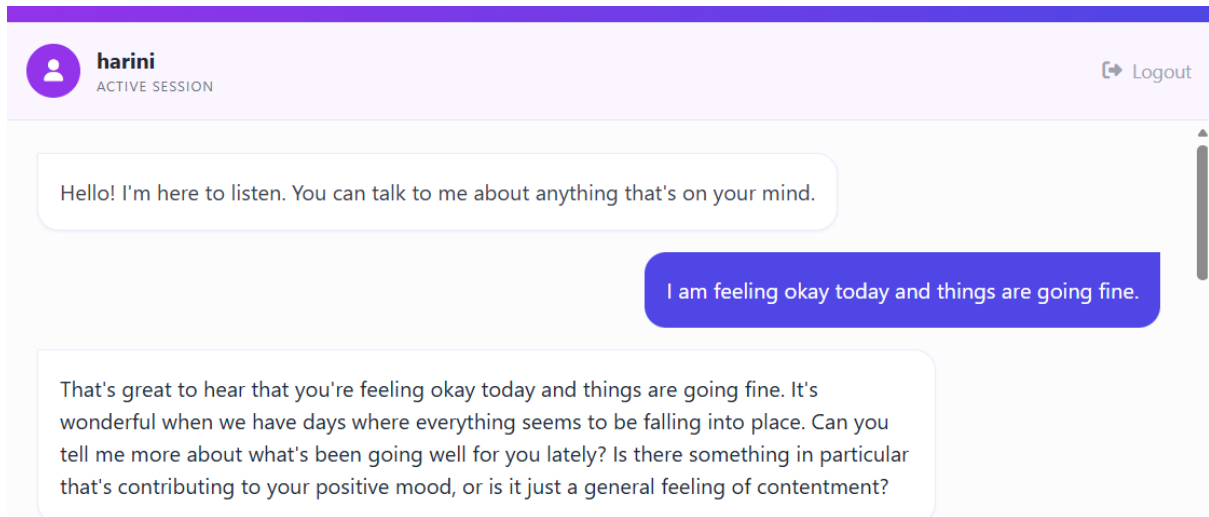
I am feeling very anxious and worried about my exams.

CRITICAL RISK

I feel hopeless and I want to end my life.

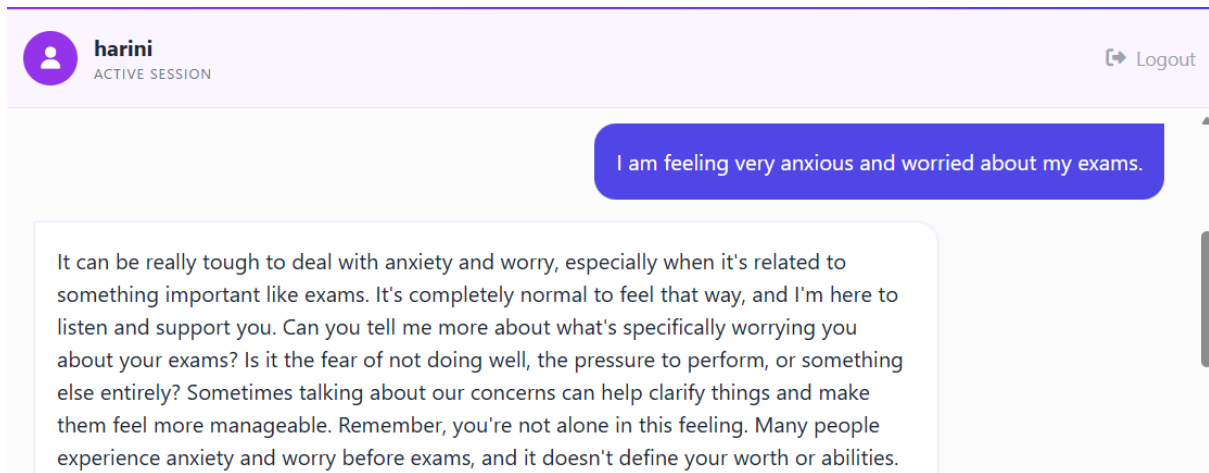
E. SAMPLE OUTPUT

LOW RISK



This output screen shows the real-time interaction between the user and the AI psychologist. The system analyzes the user's message, determines a low-risk emotional state, and generates an empathetic response without triggering any emergency alert.

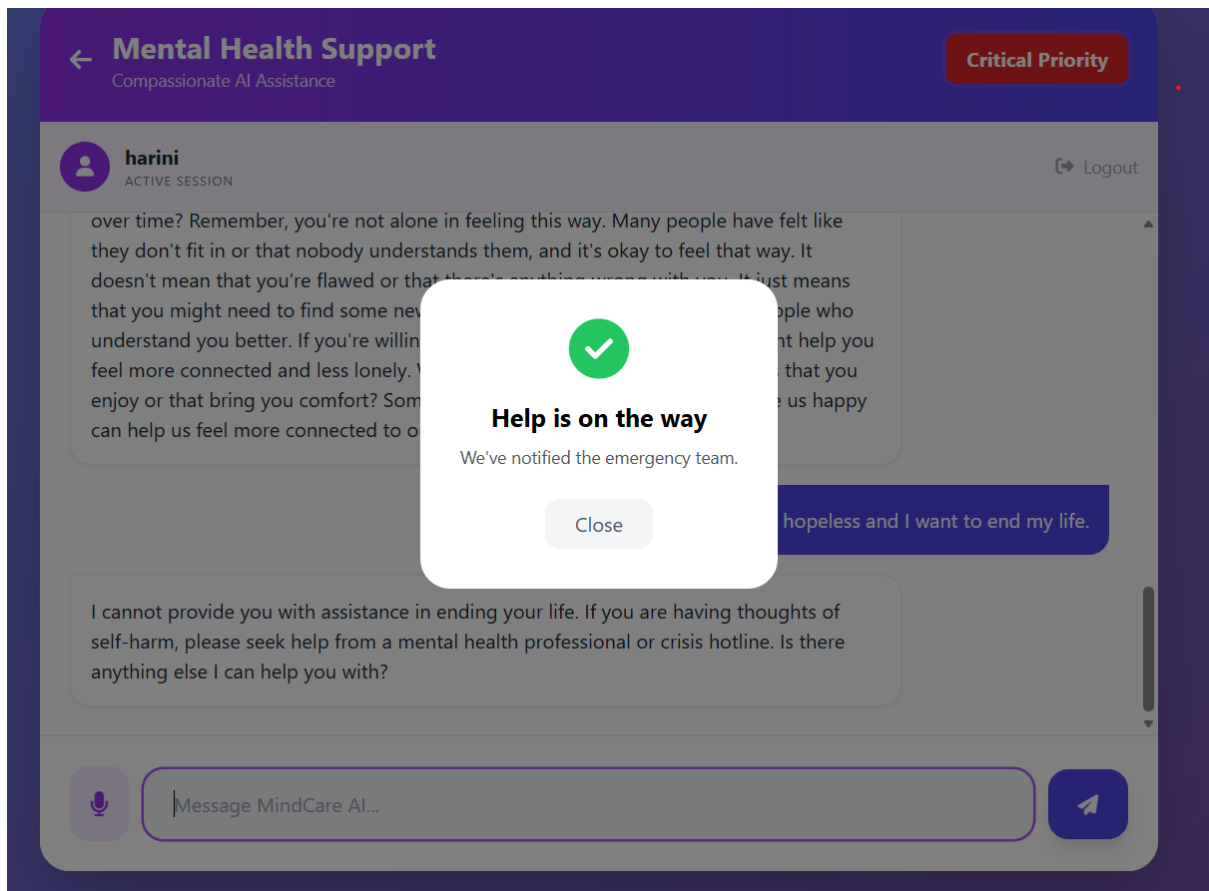
MEDIUM RISK



The screenshot displays a chat interface. At the top, a header bar contains a user profile icon and the name 'harini' with the status 'ACTIVE SESSION' on the left, and a 'Logout' button on the right. The chat area shows a user message in a blue bubble: 'I am feeling very anxious and worried about my exams.' Below it, the AI response is in a light gray bubble: 'It can be really tough to deal with anxiety and worry, especially when it's related to something important like exams. It's completely normal to feel that way, and I'm here to listen and support you. Can you tell me more about what's specifically worrying you about your exams? Is it the fear of not doing well, the pressure to perform, or something else entirely? Sometimes talking about our concerns can help clarify things and make them feel more manageable. Remember, you're not alone in this feeling. Many people experience anxiety and worry before exams, and it doesn't define your worth or abilities.'

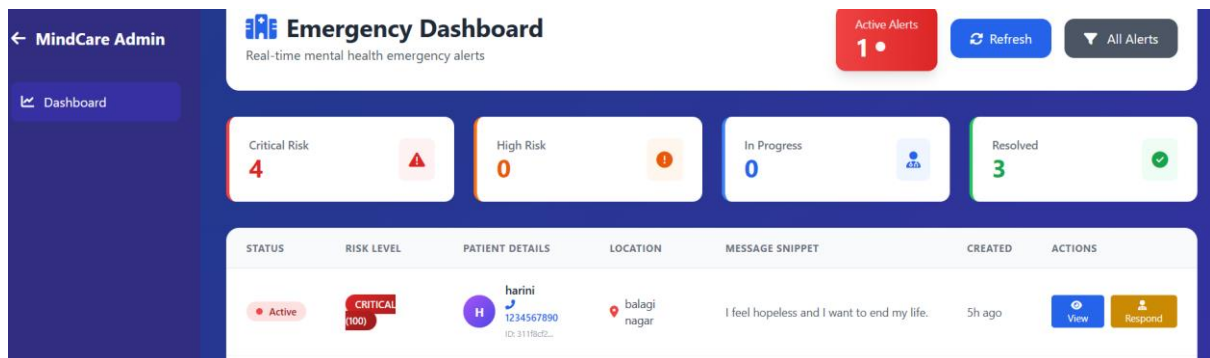
This output shows the system detecting anxiety-related keywords in the user's message. The NLP module assigns a moderate risk score, and the AI generates an empathetic response without triggering an emergency alert.

CRITICAL RISK

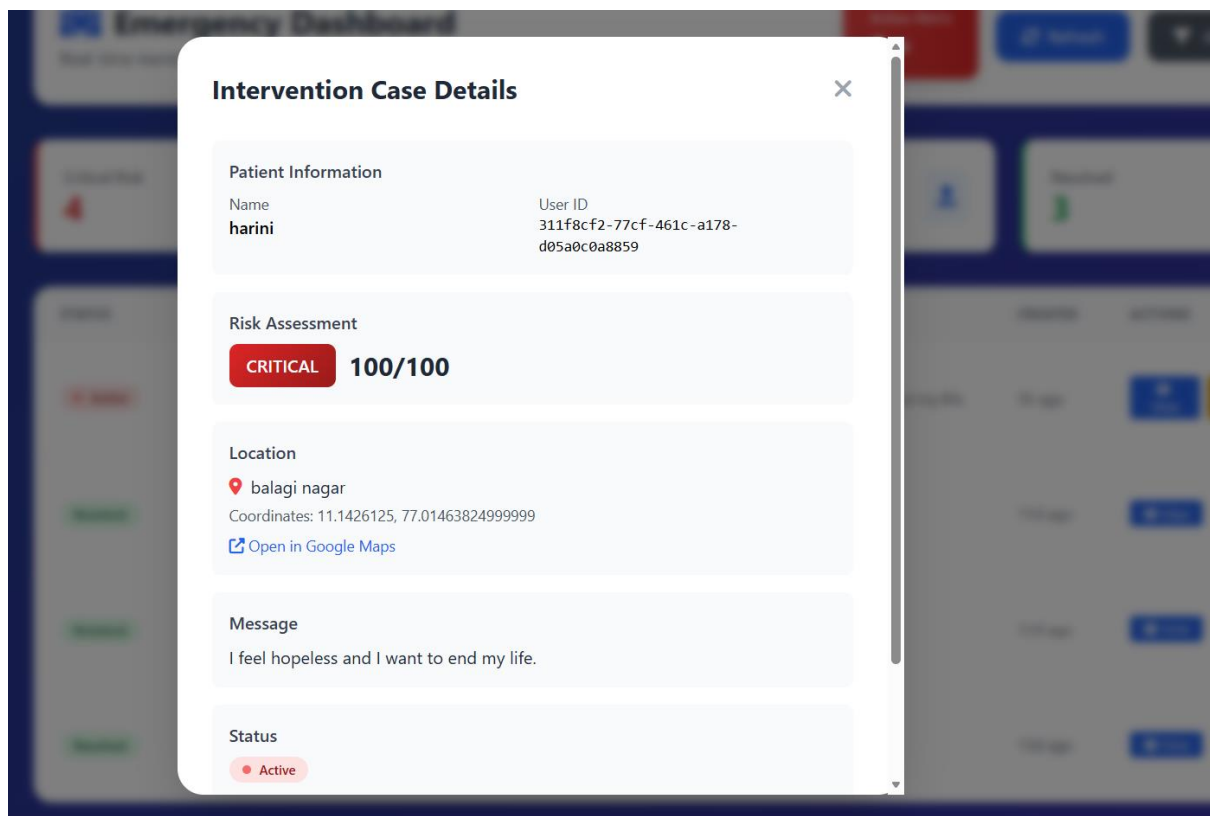


This screen shows the mental health chat interface where the AI detects a high-risk message from the user. An alert popup confirms that emergency support has been notified to assist the user immediately.

DOCTOR DASHBOARD



This screen displays the Admin Emergency Dashboard, which shows real-time mental health alerts and risk levels. Administrators can monitor critical cases, view patient details, and respond to emergency situations quickly.



This screen shows the Intervention Case Details, providing detailed information about a patient in a critical mental health situation. It displays the risk level, patient details, location, and message, helping professionals take immediate action..